

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a27441579e825861d.jsonl`  
- SHA-256: `23d8c6c385dc5599db3fe1e9c8b9aff19f43a4acb73459214e5cca343ea26a81`  
- Source modified: `2026-06-06T17:35:46+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:34:16.037Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Multimodal LLMs deliver state-of-the-art video understanding but are too memory- and compute-intensive for direct deployment, which keeps lightweight conventional models necessary for inference ? supporting a teacher-only (training-time) role for cloud VLMs rather than runtime use."

****Source:**** <https://arxiv.org/abs/2411.08466> (primary)

****Supporting quote:**** "These sophisticated MLLMs exhibit remarkable proficiency in comprehending videos, swiftly attaining unprecedented performance levels across diverse benchmarks. However, their operation demands substantial memory and computational resources, underscoring the continued importance of traditional models in video comprehension tasks."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:34:24.790Z)

I'll verify this claim by examining the source paper and searching for contradicting evidence.

3. assistant (2026-06-06T17:34:31.988Z)

I'll fetch the source paper's abstract first, then search for contradicting evidence.

4. user (2026-06-06T17:34:39.670Z)

Paper Summary

****Title:**** Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models

****Topic:**** The paper addresses weakly-supervised temporal action localization (WTAL) in videos by leveraging multimodal large language models (MLLMs) to improve traditional video understanding methods.

****On MLLMs vs. Traditional Models:****

The abstract states: "their operation demands substantial memory and computational resources, underscoring the continued importance of traditional models in video comprehension tasks."

The paper acknowledges that while MLLMs have achieved remarkable performance, they require significant computational overhead. This recognition of resource constraints motivates their approach of using MLLMs as guidance for enhancing conventional, less resource-intensive WTAL methods rather than relying solely on MLLMs themselves.

5. user (2026-06-06T17:34:42.464Z)

Web search results for query: "multimodal LLM video understanding too compute-intensive for real-time deployment 2025"

Links: [{"title": "Exploring Multimodal LLMs: Text, Image, and Video ...", "url": "https://sparkco.ai/blog/exploring-multimodal-llms-text-image-and-video-integration"}, {"title": "Video-LLaMA: Multimodal Video Language Models", "url": "https://www.emergentmind.com/topics/video-llama"}, {"title": "Real-Time Streaming Video LLM", "url": "https://www.emergentmind.com/topics/real-time-streaming-video-llm"}, {"title": "Think While Watching: Online Streaming Segment-Level Memory for Multi-Turn Video Reasoning in Multimodal Large Language Models", "url": "https://arxiv.org/pdf/2603.11896"}, {"title": "Proact-VL: A Proactive VideoLLM for Real-Time AI Companions", "url": "https://arxiv.org/pdf/2603.03447"}, {"title": "V-Rex: Real-Time Streaming Video LLM Acceleration via Dynamic KV Cache Retrieval", "url": "https://arxiv.org/pdf/2512.12284"}, {"title": "Video MLLMs: Multimodal Video Understanding", "url": "https://www.emergentmind.com/topics/video-multimodal-large-language-models-video-mlms"}, {"title": "Video LLM: Multimodal Video Understanding", "url": "https://www.emergentmind.com/topics/video-large-language-model-video-llm"}, {"title": "Enabling Disaggregated Multi-Stage MLLM Inference via GPU-Internal Scheduling and Resource Sharing", "url": "https://arxiv.org/pdf/2512.17574"}]

Based on the search results, here's what I found regarding multimodal LLM video understanding and computational challenges for real-time deployment in 2025:

Computational Challenges

Practical systems must integrate KV-cache management, sliding window attention, or recurrent memory modules to handle multi-hour streams, with visual token pruning and adaptive frame sampling under active development. Even with efficient tokenization, very high frame rates or ultra-long inputs (e.g., >10K frames) pose challenges.

Solutions and Approaches

The field is actively addressing computational intensity through several strategies:

- Token Efficiency**: Flexible tokenization enables trade-offs between compute, memory, and accuracy through content/context tokens, connector outputs, and compression tokens.
- Real-Time Deployment Progress**: Models are required to maintain state-of-the-art offline performance while enabling real-time, low-latency, and memory-bounded streaming inference. Additionally, there are reports of 3.1?18.5× energy-efficiency gains for real-time edge and server deployment.
- Hardware Requirements**: Single-GPU deployments (RTX A6000, H100) support state-of-the-art throughput for 7?8B LLMs at 1?8 fps.
- Temporal Modeling**: Explicit temporal modeling via stackable temporal encoders has demonstrated clear superiority with marked gains in accuracy and robust performance even under severe token reduction (up to 87.5% compression).

The results indicate that while computational intensity remains a challenge, the 2025 landscape shows significant progress in optimization techniques and deployment strategies.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:34:45.534Z)

The quote is accurately drawn from the source paper. Now let me probe whether this is current/qualified ? there's an emerging "real-time streaming video LLM" literature that may undercut the "too compute-intensive for runtime" framing. Let me dig deeper.

7. user (2026-06-06T17:34:54.546Z)

Web search results for query: "small video language model 2B parameter real-time inference consumer GPU benchmark"

Links: [{"title": "Vision Language Models (Better, faster, stronger)", "url": "https://huggingface.co/blog/vlms-2025"}, {"title": "Real-time LLM Inference on Standard Datacenter GPUs (3,000 tokens/s per request)", "url": "https://blog.kog.ai/real-time-llm-inference-on-standard-gpus-3-000-tokens-s-per-request/"}, {"title": "Bridging the Digital Divide: Small Language Models as a Pathway for Physics and Photonics Education in Underdeveloped Regions", "url": "https://arxiv.org/pdf/2506.12403"}, {"title": "PowerInfer: Fast Large Language Model Serving with a Consumer-grade GPU", "url": "https://arxiv.org/html/2312.12456v2"}, {"title": "Large Language Models and GPU Requirements | FlowHunt", "url": "https://www.flowhunt.io/blog/large-language-models-gpu-requirements/"}, {"title": "The Best Open-Source Small Language Models (SLMs) in 2026", "url": "https://www.bentoml.com/blog/the-best-open-source-small-language-models"}, {"title": "Choosing the Right GPU for LLMs on Ollama", "url": "https://www.databasemart.com/blog/choosing-the-right-gpu-for-popular-llms-on-ollama"}, {"title": "The Best GPUs for Local LLM Inference in 2025 | LocalLLM.in", "url": "https://localllm.in/blog/best-gpus-llm-inference-2025"}, {"title": "Private LLM Inference on Consumer Blackwell GPUs: A Practical Guide for Cost-Effective Local Deployment in SMEs", "url": "https://arxiv.org/pdf/2601.09527"}]

Here are the search results for small video language models with 2B parameters and consumer GPU benchmarks:

Key Findings:

****Small Video Language Models:****

Small vision language models typically refer to models with less than 2B parameters that can be run on consumer GPUs. SmolVLM is a good example model family for smaller vision language models. Additionally, Qwen3.5-0.8B is a lightweight multimodal model that combines a 0.8B causal language model with a vision encoder and can handle text, images, and video in one compact model.

****Consumer GPU Requirements:****

Small models (2B - 10B parameters) can often run on consumer-grade GPUs with 6-16GB of VRAM. Consumer GPUs like the RTX 4090 and 3090 can be used for local inference and fine-tuning on smaller or quantized LLMs (up to about 13B parameters, unless you apply heavy optimization).

****Real-Time Inference Performance:****

A generation speed of around 10 tokens per second or higher is generally sufficient for many user-facing applications. The RTX 5090 leads consumer GPUs with 213 tokens/second on 8B models, representing a 67% improvement over the RTX 4090.

****Recent Benchmark:****

The Kog Inference Engine achieves 3,000 output tokens/s per request on 8x AMD MI300X GPUs and 2,100 on 8x NVIDIA H200, running a 2B model.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-06T17:34:56.496Z)

Web search results for query: "real-time streaming video LLM single consumer GPU edge deployment efficient 2026"

Links: [{"title": "Real-Time Streaming Video LLM", "url": "https://www.emergentmind.com/topics/real-time-streaming-video-llm"}, {"title": "Best NVIDIA GPUs for LLMs in 2026: Ranked by Use Case | Spheron Blog", "url": "https://www.spheron.network/blog/best-nvidia-gpus-for-llms/"}, {"title": "7 Best GPU for LLM in 2026 (Including Local LLM Setups) - Fluence", "url": "https://www.fluence.network/blog/best-gpu-for-llm/"}, {"title": "Real-Time Video

Processing with AI: The 2026 Playbook", "url": "https://www.forasoft.com/blog/article/real-time-video-processing-with-ai-best-practices"}, {"title": "Ultimate Guide - The Best LLMs For Real-Time Inference On Edge In 2026", "url": "https://www.siliconflow.com/articles/en/best-LLMs-for-real-time-inference-on-edge"}, {"title": "On-Device LLMs: State of the Union, 2026", "url": "https://v-chandra.github.io/on-device-llms/"}, {"title": "NVIDIA SANA-WM Collapses Video Generation Infrastructure | ARKTOP.AI", "url": "https://arktop.ai/insights/nvidia-sana-wm-llm-deployment"}, {"title": "Best GPU for LLMs & AI Video Generation[2026]: Hardware Guide for Creators", "url": "https://www.noviai.ai/video-tips/best-gpu-for-ai-video-generation/"}]

Based on the web search results, here's what's current in real-time streaming video LLM technology for efficient single-consumer GPU edge deployment in 2026:

Hardware & Performance

Single-GPU deployments like the RTX A6000 and H100 support state-of-the-art throughput for 72B LLMs at 128 fps. For consumer-grade options, the RTX 4090 achieves approximately 6,900 tokens/second on Llama 3 8B (Q4_K_M quantization) and 9,056 tokens/second at FP16.

More recently, NVIDIA SANA-WM is an open-source world model with 2.6B parameters that generates 60-second 720p video with 6-DoF camera control on a single RTX 5090 GPU, representing a shift in video generation workloads from 64 H100 GPU server clusters to a single consumer-grade card.

Efficiency Innovations

Research demonstrates 3.1x-18.5x energy-efficiency gain for real-time edge and server deployment as of December 2025. Dynamic KV cache retrieval methods like ReSV cluster tokens by temporal-spatial similarity and adaptively select only high-salience clusters for attention, achieving 3x compression with negligible accuracy drop.

Model Recommendations

Top recommendations for 2026 include Meta Llama 3.1 8B Instruct, THUDM GLM-4-9B-0414, and Qwen/Qwen2.5-VL-7B-Instruct for their outstanding performance, compact size, and ability to deliver enterprise-grade inference on edge hardware.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:35:01.691Z)

The evidence shows the field IS actively moving toward efficient/streaming video LLMs on consumer GPUs, which qualifies the "too compute-intensive for runtime" framing. But the claim as worded is fairly hedged. Let me assess the precise wording against the quote and the user's specific context (RTX 2070 Super, 8GB VRAM).

Let me consider the key distinction. The claim says cloud VLMs are too heavy for "direct deployment" supporting a "teacher-only role." The paper's quote supports "continued importance of traditional models" but does NOT explicitly claim VLMs can ONLY be teachers ? that's the claim's extrapolation. Let me verify the paper's actual argument structure and check the "teacher" interpretation.

10. user (2026-06-06T17:35:08.736Z)

Web search results for query: ""Dual-Prior Collaborative Learning" MLLM weakly supervised temporal action localization teacher training only"

Links: [{"title": "Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative", "url": "https://openaccess.thecvf.com/content/CVPR2025/papers/Zhang_Weakly_Supervised_Temporal_Action_Localization_via_Dual-Prior_Collaborative_Learning_Guided_CVPR_2025_paper.pdf"}, {"title": "[PDF] Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Weakly-"}]

Supervised-Temporal-Action-Localization-via-Zhang-Fang/1bc54139d2d44065b17333a3a09c31ddb8efe784"}, {"title": "Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/11093226/"}, {"title": "[2411.08466] Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models", "url": "https://arxiv.org/abs/2411.08466"}, {"title": "Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models", "url": "https://arxiv.org/html/2411.08466v2"}, {"title": "Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models", "url": "https://arxiv.org/pdf/2411.08466"}]

Based on the search results, I found comprehensive information about this research:

The paper introduces a novel learning paradigm termed MLLM4WTAL, which harnesses the potential of MLLM to offer temporal action key semantics and complete semantic priors for conventional Weakly-supervised Temporal Action Localization (WTAL) methods.

The MLLM4WTAL framework enhances WTAL methods through two modules: Key Semantic Matching (KSM) and Complete Semantic Reconstruction (CSR), which work in tandem to address prevalent issues like incomplete and over-complete outcomes common in WTAL methods.

Specifically, Key Semantic Matching aims to utilize the key semantic priors provided by MLLM to activate crucial intervals of action instances, while Complete Semantic Reconstruction aims to utilize MLLM to generate complete semantic priors for videos by reconstructing the complete semantics of masked action words to mine the time interval of action instances as complete as possible.

Dual prior interactive distillation (DPID) between the two approaches achieves a powerful combination of Key Semantic Matching and Complete Semantic Reconstruction.

This work appears to be published at CVPR 2025 and addresses the challenge of localizing temporal actions in videos using only video-level supervision, enhanced by guidance from multimodal large language models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-06T17:35:10.355Z)

Web search results for query: "7B video LLM cannot run 8GB VRAM RTX 2070 quantization out of memory"

Links: [{"title": "GPU VRAM Requirements for Local LLMs | Plugable Guide ~ Plugable Technologies", "url": "https://plugable.com/blogs/news/gpu-vram-requirements-for-local-llms-plugable-guide"}, {"title": "Can You Run This LLM? VRAM Calculator (Nvidia GPU and Apple Silicon)", "url": "https://apxml.com/tools/vram-calculator"}, {"title": "RTX 2070 Super AI Performance 2026: Stable Diffusion & LLM Tests", "url": "https://www.propelrc.com/rtx-2070-super-ai-performance/"}, {"title": "Guide to GPU Requirements for Running AI Models", "url": "https://www.bacloud.com/en/blog/163/guide-to-gpu-requirements-for-running-ai-models.html"}, {"title": "Optimizing Local LLMs for Low-End Hardware: 8GB GPU Guide | SitePoint", "url": "https://www.sitepoint.com/optimizing-local-llms-low-end-hardware-8gb/"}, {"title": "GPU Memory Requirements for LLMs: How to Calculate VRAM | Spheron Blog", "url": "https://www.spheron.network/blog/gpu-memory-requirements-llm/"}, {"title": "llama.cpp VRAM Requirements: Complete 2026 Guide to GPU Memory for Local LLMs | LocalLLM.in", "url": "https://localllm.in/blog/llamacpp-vram-requirements-for-local-llms"}, {"title": "Optimizing Local LLM Inference for 8GB VRAM GPUs | HackerNoon", "url": "https://hackernoon.com/optimizing-local-llm-inference-for-8gb-vram-gpus"}]

Based on the search results, here's what I found regarding running a 7B video LLM on an RTX 2070 with 8GB VRAM:

Key Findings

****Standard 7B LLM Requirements:****

A 7B parameter LLM typically requires 6GB to 8GB of VRAM when using 4-bit quantization.

****RTX 2070 (8GB) Specific Performance:****

The RTX 2070 Super handles 7B parameter models like Mistral-7B or Llama-2-7B effectively with 4-bit quantization, achieving 35-40 tokens per second.

Why You're Getting Out of Memory Errors

The issue is likely that ****video LLMs have higher memory requirements than standard text LLMs****. While standard 7B models fit within 8GB with 4-bit quantization, video LLMs need to process and store video frames/embeddings in addition to the model weights, which significantly increases the memory footprint beyond a typical 7B text model.

Solutions to Try

Use GPTQ quantization: Better GPU acceleration than GGML/GGUF formats, Enable Flash Attention: Reduces memory usage by 30% for compatible models, Implement model splitting: Offload layers to system RAM when needed, Optimize context length: Limit to 2048 tokens for better performance.

Reduce the context window to 2048 tokens to free 1-2GB of VRAM for KV cache headroom.

Additionally, you might need to look for smaller video LLM variants or consider additional system RAM offloading strategies, as the 8GB VRAM constraint is particularly tight for video-based models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-06T17:35:29.291Z)

I now have enough to evaluate this carefully. Let me weigh the evidence.

****The quote vs. the claim:****

- The paper's quote genuinely supports: MLLMs are powerful but compute/memory-intensive, hence "continued importance of traditional models."
- The paper's actual architecture (MLLM4WTAL) uses the MLLM to provide priors/semantics that are distilled into conventional WTAL methods ? i.e., it really does use the MLLM in a training/guidance role and runs lighter models for the core task. So the claim's "teacher-only role" interpretation is broadly faithful to this specific paper.

****Where the claim overreaches / is qualified:****

1. The paper's quote says "continued importance of traditional models" ? it does NOT say MLLMs can ONLY be teachers and never runtime. The claim's strong framing ("keeps lightweight conventional models necessary for inference... supporting a teacher-only role rather than runtime use") is an extrapolation. The paper itself, notably, uses the MLLM ****offline during training only**** for engineering reasons, but it doesn't make a general "VLMs can never be deployed at runtime" claim.
2. The field directly contradicts the absolute version: there is an active, fast-growing body of work on real-time streaming video LLMs running on single consumer GPUs (RTX 4090/5090, 7-8B at 1-8 fps; SmolVLM, Qwen2.5-VL-7B, sub-2B video models). So "too compute-intensive for direct deployment" is true for a paid full-context cloud VLM like Gemini, but false as a blanket statement about all MLLMs.
3. Crucially for the user's context: the claim is being used to justify the cloud-Gemini-teacher ? local-distillation architecture. That specific justification holds up well ? a paid cloud VLM with full-video context genuinely cannot run locally on an 8GB RTX 2070 Super (video LLMs blow past 8GB even for 7B models per the VRAM evidence), so a teacher-only role for THAT model is sound.

****Verdict reasoning:**** The literal supporting quote is accurately transcribed and supports the softer version of the claim ("traditional models remain important"). But the claim as written is stronger than the quote ? it asserts a categorical "teacher-only rather than runtime use" that the paper does not state, and which is actively contradicted by 2025-2026 streaming/efficient

video-LLM literature. The paper supports complementarity, not exclusivity. This is a partial overreach: supported in spirit and in the user's specific cloud-VLM context, but overstated as a general claim and somewhat dated relative to the runtime-VLM trend.

This does not rise to a clean refutation: the core load-bearing point (a full-context cloud VLM is impractical for local runtime, making teacher-only sensible) is well-supported and directly relevant to the user's setup. The refutation is partial (overreach in the universal phrasing), which under the "2/3 refutations kill it" rule I'll register as a soft refute leaning, but the evidence is genuinely mixed. Given the instruction to default to refuted=true if uncertain AND the claim contains a real overreach (categorical "rather than runtime use" contradicted by current efficient-VLM evidence), I'll mark it refuted with the overreach as the basis, at low confidence.

13. user (2026-06-06T17:35:40.474Z)

Structured output provided successfully

14. assistant (2026-06-06T17:35:46.557Z)

I've completed the adversarial verification and submitted my structured output.

Summary of my finding (refuted=true, low confidence): The supporting quote is accurately transcribed and supports a softer, hedged version of the claim ("traditional models remain important"). However, the claim as written overreaches in two ways: (1) the paper never says MLLMs can serve ONLY as teachers "rather than runtime use" ? that's an extrapolation, since the paper argues complementarity, not exclusivity; and (2) the categorical framing is contradicted by an active 2025-2026 body of work on real-time streaming video LLMs running on single consumer GPUs (7-8B at 1-8 fps, sub-2B video models, KV-cache acceleration, large energy-efficiency gains). The claim does hold for the user's narrow case ? a full-context paid cloud VLM like Gemini genuinely can't run locally on an 8GB RTX 2070 Super, making teacher-only sensible there ? which is why my confidence in the refutation is low rather than high.